ModelFramework

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Model

Notes to myself

- Find the best predictors of juveniles densities
- Should we care predictability ** We want to predict occurrence and abundance to prioritize management action *** Changing covariates in the future.

Mixed effects models * Covariance model * The regression coefficients would be a map * Time varying coefficients

- Packages

bmrs, glmmTMB

zero-inflated negative binomial

- ##Data

Sample sizes

Table 1: Sample sizes by stratum and year.

	Lakes	Mid-Coast	Mid-South Coast	North Coast	Umpqua
1998	0	34	33	25	0
1999	0	30	26	32	28
2000	5	38	20	35	30
2001	1	39	22	63	22
2002	1	39	23	50	24
2003	1	35	25	44	34
2004	1	36	28	45	25
2005	1	42	26	45	32
2006	2	38	25	52	30
2007	5	56	36	47	43
2008	4	51	35	39	50
2009	2	43	33	31	35
2010	2	36	38	35	33
2011	2	44	34	34	37
2012	3	43	36	42	41

	Lakes	Mid-Coast	Mid-South Coast	North Coast	Umpqua
2013	2	44	35	33	37
2014	2	30	25	22	31
2015	3	38	28	25	32
2016	3	35	27	31	29
2017	2	43	30	34	35
2018	2	38	32	34	36
2019	0	39	36	34	34

##glmmTMB

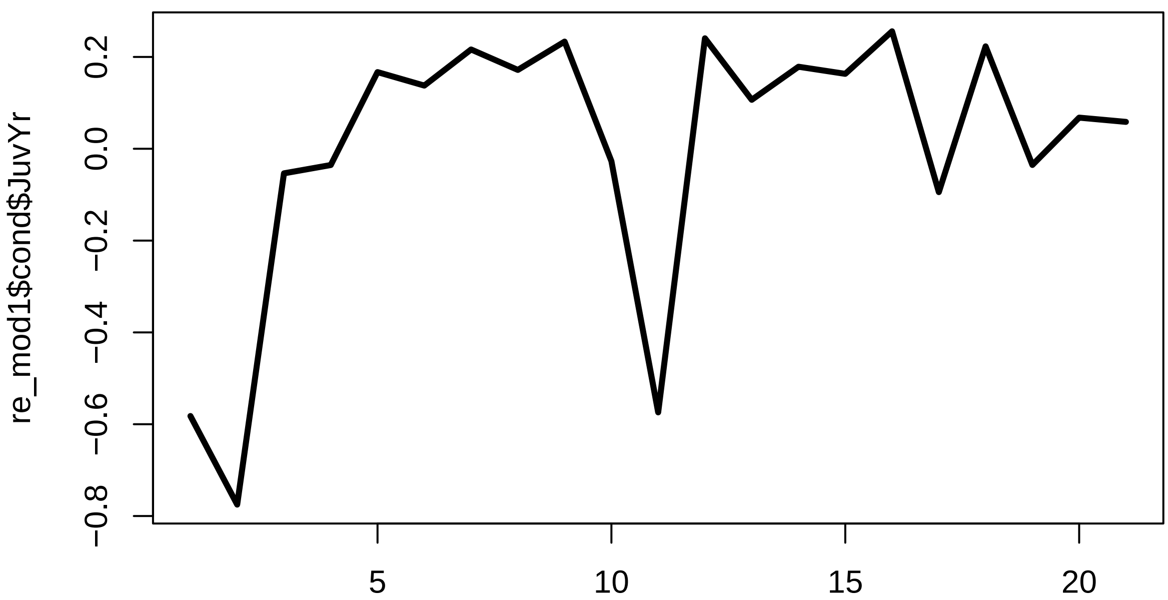
```
trainJuv <- juv[juv$JuvYr<max(juv$JuvYr),]
predJuv <- juv[juv$JuvYr==max(juv$JuvYr),]

#Stream yr independent
juv_ziNB_yr_Stratum <- glmmTMB(Juv.km~StrmSlope + MaxGradD + (1|Stratum) +(1|JuvYr),
                    data=trainJuv,
                    ziformula=~1,
                    family=nbinom1)

#Stream year and stratum, independent AR1 processes for each stream
#This ar1 notation seems very backwards to me
juv_ziNB_ar1_yr_stratum <- glmmTMB(Juv.km~StrmSlope + MaxGradD + ar1(Stratum+0|JuvYr),
                    data=trainJuv,
                    ziformula=~1,
                    family=nbinom1)

#Stream year and stratum, uncorrelated processes for each stream
#Computationally too expensive, >1 min = too long
#Distance between stratum would reduced dimensionality and speed estimation
#could use ou covariance structure
# juv_ziNB_us_yr_stratum <- glmmTMB(Juv.km~StrmSlope + MaxGradD + us(as.factor(JuvYr)+0|Stratum),
#                      data=trainJuv,
#                      # dispformula=~0,
#                      ziformula=~(1|Stratum),
#                      family=nbinom1)


pred_mod1 <- predict(juv_ziNB_yr_Stratum,
                     newdata= predJuv)
```

```
## Warning in checkTerms(data.tmb1$terms, data.tmb0$terms): Predicting new random effect levels for ter
## Disable this warning with 'allow.new.levels=TRUE'
```

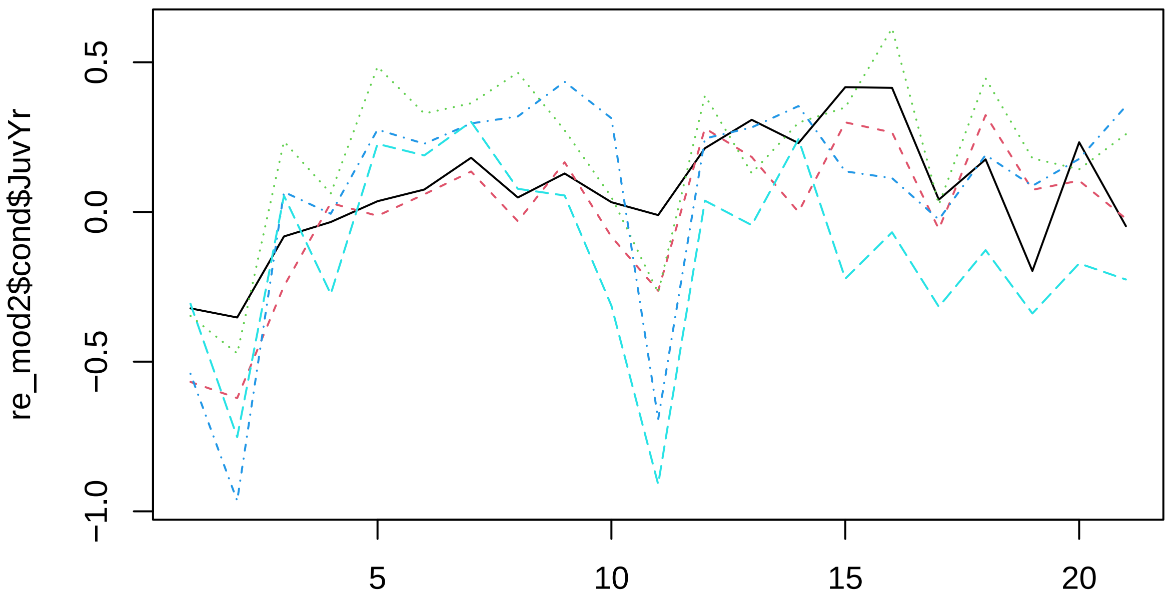
```
pred_mod2 <- predict(juv_ziNB_ar1_yr_stratum,
                     newdata= predJuv,
                     allow.new.levels = TRUE)

MAE <- data.frame(mod1=sum(abs(exp(pred_mod1)-predJuv$Juv.km)),
                  mod2=sum(abs(exp(pred_mod2)-predJuv$Juv.km)))

print(MAE)
```

```
##       mod1     mod2
## 1 45974.53 46444.69
```

```r
re_mod1 <- ranef(juv_ziNB_yr_Stratum)
re_mod2 <- ranef(juv_ziNB_ar1_yr_stratum)
matplot(re_mod1$cond$JuvYr,
        type="l",
        lwd=3)
```




```r
matplot(re_mod2$cond$JuvYr,
        type="l")
```


re_mod2$cond$JuvYr
0.5
0.0
−0.5
−1.0
5
10
15
20